

CSHLvc2026

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Welcome

Package: CSHLvc2026 **Authors:** Vincent Carey [aut, cre] **Compiled:** 2026-06-12 **Package version:** 0.98.1 **R version:** R version 4.6.0 (2026-04-24) **BioC version:** 3.24 **License:** MIT + file LICENSE

This is the landing page of the **BiocBook** entitled

This book introduces the reader to

Docker image

A Docker image built from this repository is available here:

ghcr.io/vjcitn/cshlvc2026

Get started now

You can get access to all the packages used in this book in < 1 minute, using this command in a terminal:

Listing 0.1 bash

```
docker run -it ghcr.io/vjcitn/cshlvc2026:devel R
```

RStudio Server

An RStudio Server instance can be initiated from the **Docker** image as follows:

Listing 0.2 bash

```
docker run \  
  --volume <local_folder>:<destination_folder> \  
  -e PASSWORD=OHCA \  
  -p 8787:8787 \  
  ghcr.io/vjcitn/cshlvc2026:devel
```

The initiated RStudio Server instance will be available at <https://localhost:8787>.

Session info

 Click to expand

1 Introduction to GWAS

This chapter introduces genome-wide association studies (GWAS), covering the conceptual foundations and practical tools needed to design, execute, and interpret large-scale genetic association analyses.

1.1 Phenotype Vocabulary

At the boundaries of knowledge, it is important to establish conventions to support clear communication. “Ontology” is a field of information science devoted to formal specification of vocabularies and associated conceptual hierarchies. Practical use of ontologies in genomic data science started with Gene Ontology but has grown to involve frequent use of ontologies for

- cell types
- diseases
- experimental factors and designs

Here is an “ontological report” on “exocrine pancreatic carcinoma”, provided by the European Bioinformatics Institute Ontology Lookup Service:

exocrine pancreatic carcinoma

http://purl.obolibrary.org/obo/MONDO_0005192 [Copy](#)

Definition:
A carcinoma that arises from epithelial cells of the exocrine pancreas ⓘ

Also appears in CPONT EFO OBA PRIDE

Exact Synonyms carcinoma of exocrine pancreas ⓘ carcinoma of the pancreas ⓘ exocrine pancreas carcinoma ⓘ pancreas carcinoma ⓘ pancreatic cancer (not islets) ⓘ

Related Synonyms pancreatic carcinoma, familial ⓘ

Broad Synonyms cancer of pancreas ⓘ cancer of the pancreas ⓘ exocrine cancer ⓘ pancreas cancer ⓘ pancreatic cancer ⓘ

Search MONDO... [Search](#)

☐ Exact match ☐ Include obsolete terms ☒ Include imported terms Search Model: Lexical ▼

Tree **Graph**

disease (29,314)
 human disease (23,213)
 disease by body system or component (18,799)
 digestive system disorder (1,428)
 digestive system cancer (450)
 digestive system carcinoma (281)
 exocrine pancreatic carcinoma (26)
 malignant pancreatic neoplasm (37)
 malignant exocrine pancreas neoplasm (31)
 exocrine pancreatic carcinoma (26)
 pancreas disorder (184)
 pancreatic neoplasm (71)
 malignant pancreatic neoplasm (37)
 malignant exocrine pancreas neoplasm (31)
 exocrine pancreatic carcinoma (26)

☒ Preferred roots
☐ All classes
☒ Show counts
☐ Show obsolete terms
☐ Show all siblings

Class Information

Conforms To
<http://purl.obolibrary.org/obo/mondo/patterns/carcinoma.yaml>

curated content resource
https://www.malacards.org/card/pancreatic_cancer ⓘ

database_cross_reference

- pancreatic carcinoma [OBO](#) ⓘ
- obsolete_pancreatic carcinoma [EFO](#) ⓘ
- GARD:0027717 ⓘ
- MEDGEN:65917 ⓘ
- Exocrine Pancreas Carcinoma [NCIT](#) ⓘ
- SCTID:372142002 ⓘ
- UMLS:C0235974 ⓘ

Figure 1.1: mondo snapshot

Key elements:

- definition of the short phrase or “term”
- “IRI”: a URL-like entity that can be used for web access
- “CURIE”: compact universal resource identifier of fixed length (MONDO:0005192)
- snapshot of conceptual hierarchy “tree”

Let’s program with the MONDO ontology. We need to have the devtools and remotes packages installed, and then we can install the ontoProc2 package with `BiocManager::install("vjcitn/ontoProc2")`. Once this is done, the following will succeed.

```
library(ontoProc2)
## OpenTelemetry error: there is no package called 'otelSdk'
## OpenTelemetry error: there is no package called 'otelSdk'
mondo = semsql_connect(ontology="mondo") # download on first attempt
## Connected to SemanticSQL database: /root/.cache/R/BiocFileCache/18f79e7474a_mondo.db
## Primary ontology prefix: MONDO
report(mondo)
##
## =====
```

```

## SemsqConn Object
## =====
##
## Connection Details:
## -----
##   Database path:    /root/.cache/R/BiocFileCache/18f79e7474a_mondo.db
##   Ontology prefix: MONDO
##   Status:           Connected
##
## Database Statistics:
## -----
##   Labeled terms:    59,064
##   Direct edges:     109,103
##   Entailed edges:   5,003,896
##   Definitions:      22,594
##
## Terms by Prefix (top 5):
## -----
##   MONDO:             31,886
##   UBERON:             5,359
##   HGNC:               5,023
##   HP:                 4,837
##   GO:                 4,072
##
## Key Tables Available:
## -----
##   rdfs_label_statement
##   has_text_definition_statement
##   edge
##   entailed_edge
##   rdfs_subclass_of_statement
##   owl_some_values_from
##   has_oio_synonym_statement
##
## =====
## Use methods like search_labels(), get_ancestors(), etc.
## Run ?SemsqConn for documentation.
## =====

```

1.2 Study Designs

GWAS can be conducted under several study designs, each with distinct strengths and confounders. This section compares case-control studies, cohort studies, family-based designs, and biobank-scale analyses, with attention to population stratification and the importance of ancestry-matched controls.

1.3 Manhattan Plots

The Manhattan plot is the canonical visualization for GWAS results, displaying $-\log_{10}(p)$ values across the genome. This section explains how to interpret Manhattan plots, choose significance thresholds (genome-wide significance at $p < 5 \times 10^{-8}$), and identify credible association signals amid linkage disequilibrium patterns.

1.4 Functional Interpretation

Identifying a significant SNP is only the beginning. This section covers post-GWAS functional annotation — including regulatory element overlap, eQTL colocalization, gene prioritization strategies, and pathway enrichment — to move from statistical association to biological insight.

2 xQTL

This chapter covers expression and molecular quantitative trait loci (xQTL), connecting genetic variation to intermediate molecular phenotypes as a bridge between GWAS signals and biological mechanisms.

3 Spatial Transcriptomics

This chapter introduces spatial transcriptomics technologies and analytical frameworks, exploring how the spatial context of gene expression enriches our understanding of tissue architecture and disease.

4 Future Computing

This chapter looks ahead at emerging computational paradigms relevant to genomics and biomedical research, including cloud-native workflows, federated analysis, and the role of AI/ML in large-scale data integration.